PROFESSIONAL ETHICAL HACKING TRACK · PEHT

MODULE 03

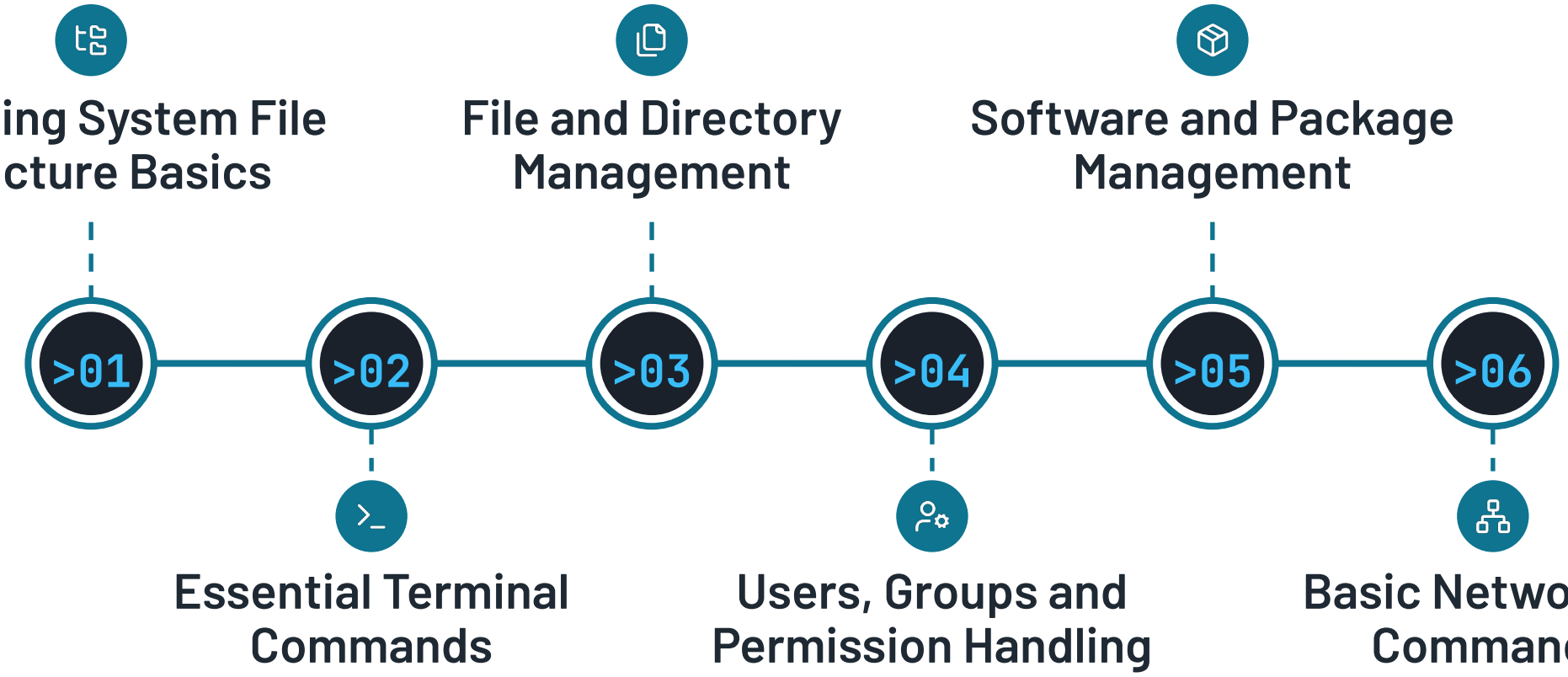
Command Line Mastery for Security Testing

INSTRUCTOR

Mo'tasem Abu Baha'

CYBERA4U

Six lessons, one terminal workflow



Why do security pros live in the terminal?

GUI

- Click by click
- Manual repeats
- Needs a desktop
- Few tools

	GUI	CLI
Speed	Click by click	One line
Automation	Manual repeats	Scriptable
Remote access	Needs a desktop	Works over SSH
Tool availability	Few tools	Every tool

CLI

- One line
- Scriptable
- Works over SSH
- Every tool

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By the end of this module, you will...



>01

Navigate the Linux filesystem



>02

Read and filter output



>03

Manage files, users and permissions



>04

Install tools and run services



>05

Inspect your lab network

LESSON 01

Operating System File Structure Basics

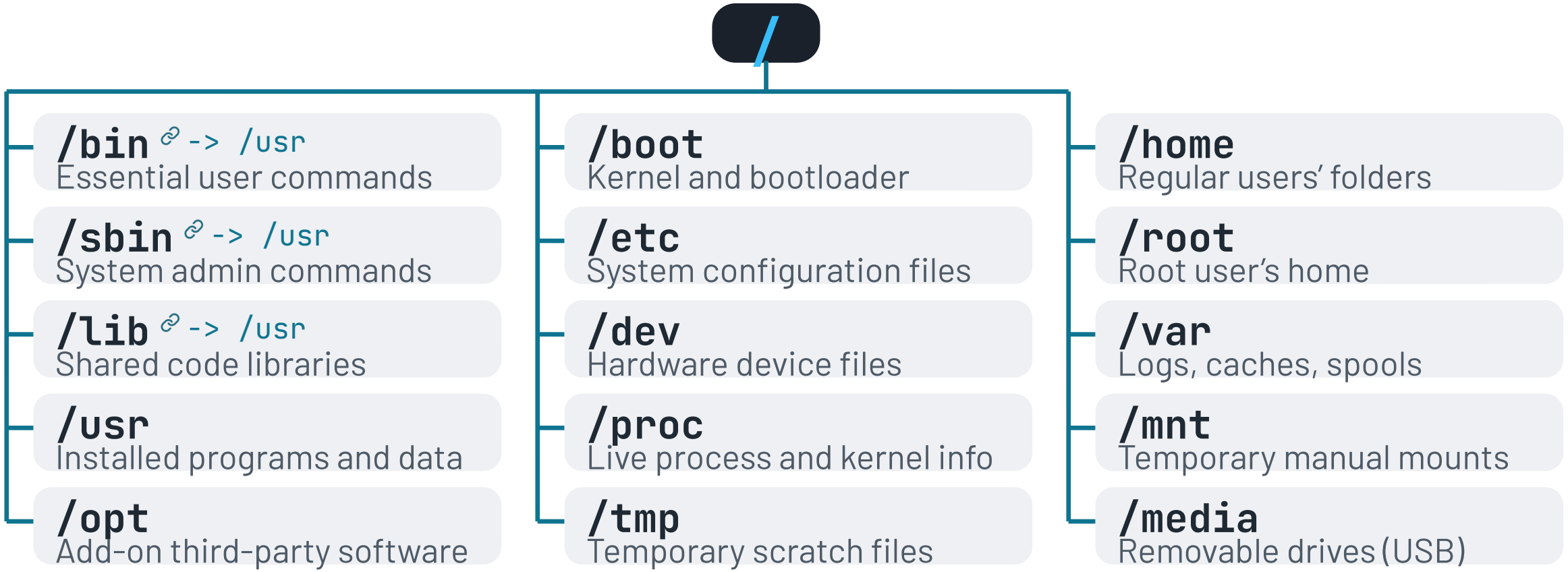
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In Linux, everything is a file



CYBERA4U

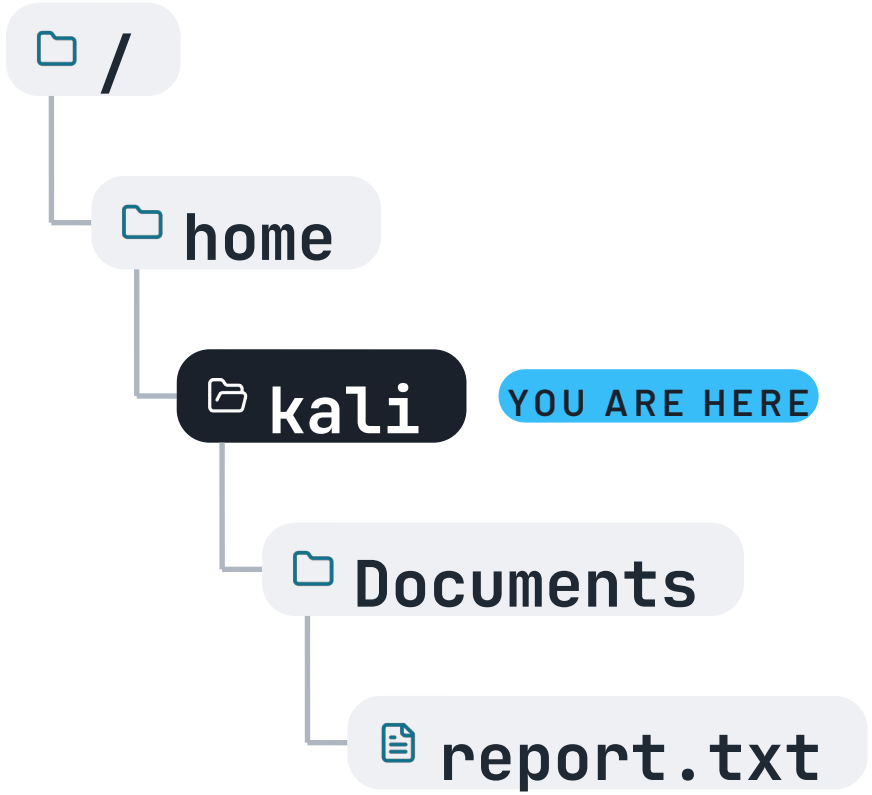
Where does everything live?



Kali uses usrmerge: /bin, /sbin and /lib link into /usr `/bin -> usr/bin`

CYBERA4U

Absolute or relative: where does the path start?



ABSOLUTE · STARTS AT /

`/home/kali/Documents/report.txt`

Works from any directory

RELATIVE · STARTS WHERE YOU ARE

`Documents/report.txt`

Depends on your current directory

```
kali@kali: ~
┌──(kali㉿kali)-[~]
└─$ pwd
/home/kali
```


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What do ~, . and .. stand for?

~ **Your home**
/home/kali

. **This directory**
where you are

.. **Parent directory**
one level up

kali@kali: ~

```
┌──(kali㉿kali)-[~]
└─$ ls -a
.               .bashrc.original  Documents  .local     Public
..              .cache            Downloads  Music      Templates
.bash_logout    .config           .face      Pictures   Videos
.bashrc         Desktop           .face.icon .profile   .zshrc
```

Names starting with `.` are hidden; `ls -a` reveals them

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Linux vs. Windows: what changes?

	Linux (Kali)	Windows
Top of the tree	`/`	`C:\  D:\`
Path separator	`/`	`\`
Letter case	Case-sensitive	Case-insensitive
System configuration	`/etc`	`C:\Windows\System32`
User folders	`/home`	`C:\Users`

`Notes.txt ≠ notes.txt` Two different files on Linux

CYBERA4U

Which folders will you open every day?

/etc
System configuration

`/etc/passwd` `/etc/hosts`

/var/log
Logs and system history

`/var/log/dpkg.log` `/var/log/apt/`

/home /root
User data and settings

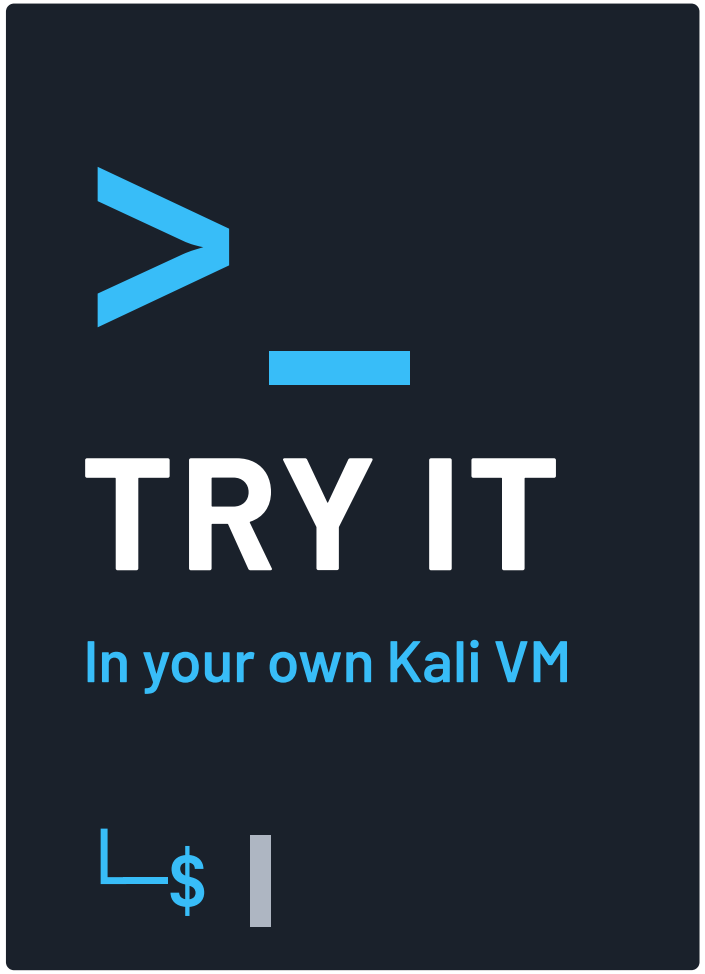
`/home/kali` `/root`

/usr/share
Shared data, incl. Kali wordlists

`/usr/share/wordlists` `/usr/share/nmap`

CYBERA4U

Try it: explore the filesystem



>01 Print your current directory.

>02 Prove that /bin is a symlink.

>03 Reveal the hidden files in your home.

>04 List Kali's wordlists by absolute path.

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What to remember from Lesson 01

+01



One tree, rooted at /

+02



Absolute starts at /; relative starts here

+03



Know /etc, /var/log, /home, /usr/share

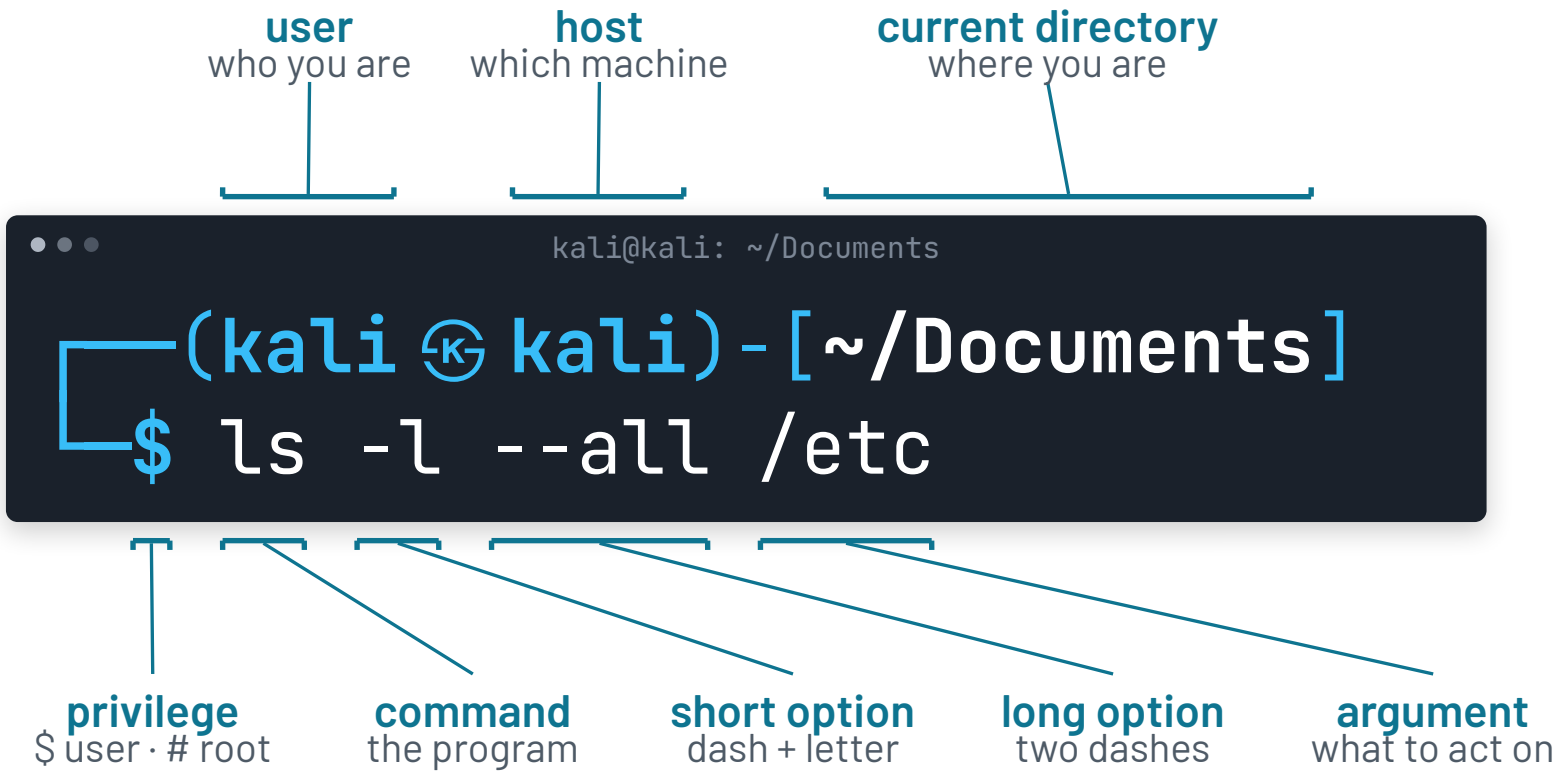
CYBERA4U

LESSON 02

Essential Terminal Commands

CYBERA4U

How do you read a command line?



Root shell looks different



⚠ # = full system power
Elevate only when needed

Stuck? Ask the system first

ONE LINE
`whatis ls`
Short summary

→

ONE SCREEN
`ls --help`
Usage and options

→

FULL MANUAL
`man ls`
Everything · q quits

SEARCH
`apropos copy`
Search by keyword

kali@kali: ~

```
┌──(kali㉿kali)-[~]
└─$ whatis ls
ls (1)                   - list directory contents
```

kali@kali: ~

```
┌──(kali㉿kali)-[~]
└─$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILEs (the current directory by default).
```

Who am I, and which box is this?

kali@kali: ~

```
┌──(kali㉿kali)-[~]
└─$ whoami
kali
┌──(kali㉿kali)-[~]
└─$ hostname
kali
```

kali@kali: ~

```
┌──(kali㉿kali)-[~]
└─$ id
uid=1000(kali) gid=1000(kali) groups=1000(kali),
4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),
29(audio),30(dip),44(video),46(plugdev),100(users),…
```

kali@kali: ~

```
┌──(kali㉿kali)-[~]
└─$ pwd
/home/kali
```

kali@kali: ~

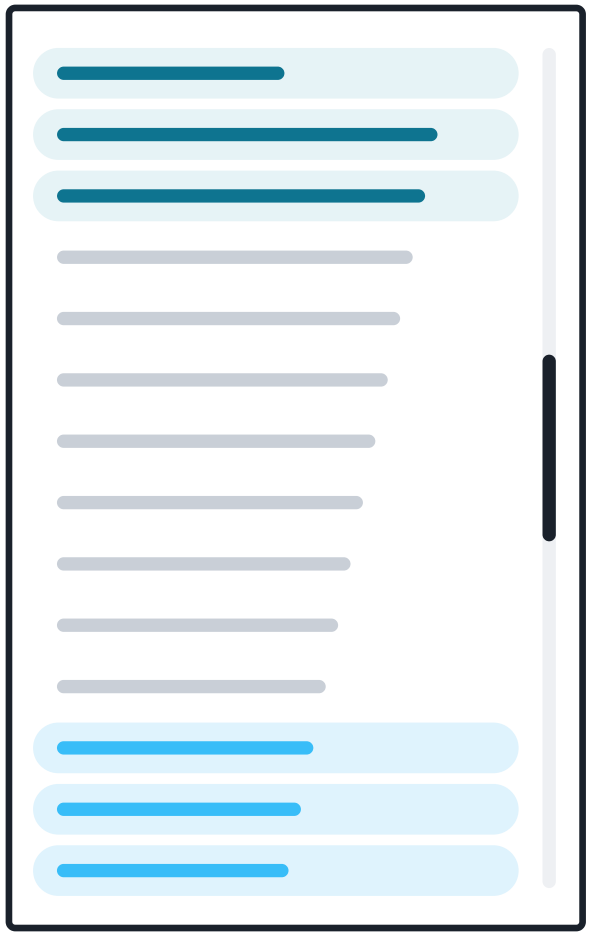
```
┌──(kali㉿kali)-[~]
└─$ uname -a
Linux kali 6.12.25-amd64 #1 SMP PREEMPT_DYNAMIC
Kali 6.12.25-1kali1 (2025-04-30) x86_64 GNU/Linux
```

whoami · hostname · pwd
who, which machine, where

id · uname -a
privileges, kernel, architecture

CYBERA4U

How do you read a file without an editor?



head

less

tail

cat = whole file

`cat file`	Print the whole file
`less file`	Scroll and search · q quits
`head -n 5 file`	First 5 lines
`tail -f log`	Follow new lines live

```
kali@kali: ~

┌──(kali㉿kali)-[~]
└─$ head -n 3 /etc/passwd
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
```


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Filter output down to what matters

`grep` Find lines that match

- `-i` ignore case
- `-r` search folders
- `-v` invert match
- `-n` line numbers

```
kali@kali: ~
┌──(kali㉿kali)-[~]
└─$ grep -i pretty_name /etc/os-release
PRETTY_NAME="Kali GNU/Linux Rolling"
```

Matches highlighted, case ignored

`sort`

Order lines

```
guest
admin
root
```

→

```
admin
guest
root
```

`uniq -c`

Count repeats

```
admin
admin
root
```

→

```
2 admin
1 root
```

`cut -d: -f1`

Pick a field

```
root:x:0:0
bin:x:2:2
```

→

```
root
bin
```


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Chain small tools into one answer



`|`	Pass output on
`>`	Overwrite a file
`>>`	Append to a file
`2>`	Errors to a file
`2>/dev/null`	Discard errors

```
kali@kali: ~
┌──(kali㉿kali)-[~]
└─$ ls /root
ls: cannot open directory '/root': Permission denied
┌──(kali㉿kali)-[~]
└─$ ls /root 2>/dev/null
```

Same command, error discarded

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Type less, recover faster

Tab	↑	Ctrl + R	Ctrl + C	Ctrl + L
Complete names	Last command	Search history	Stop a command	Clear screen

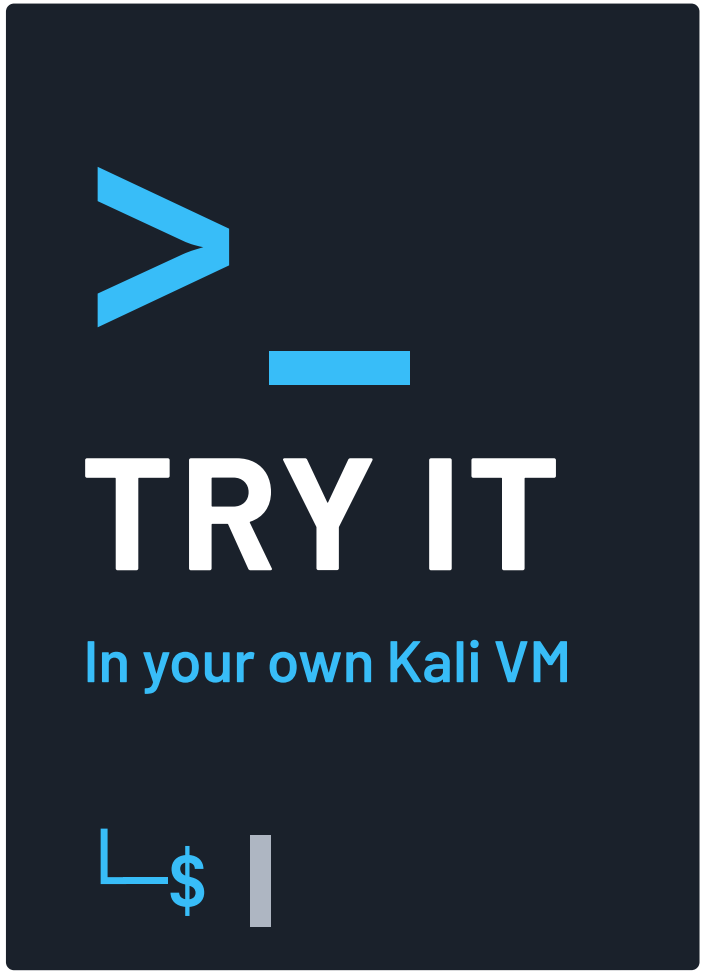
history List past commands

!! Repeat last command

Kali shows the expanded line first



Try it: read, filter, chain



>01 Get a one-line summary of grep.

>02 Show your UID and every group.

>03 Print the last 5 lines of /etc/passwd.

>04 Count accounts whose shell is nologin.

What to remember from Lesson 02

+01

Read the prompt: user, host, path, privilege

+02

Stuck? `whatis`, then `--help`, then `man`

+03

Pipe `|` small tools into one answer

LESSON 03

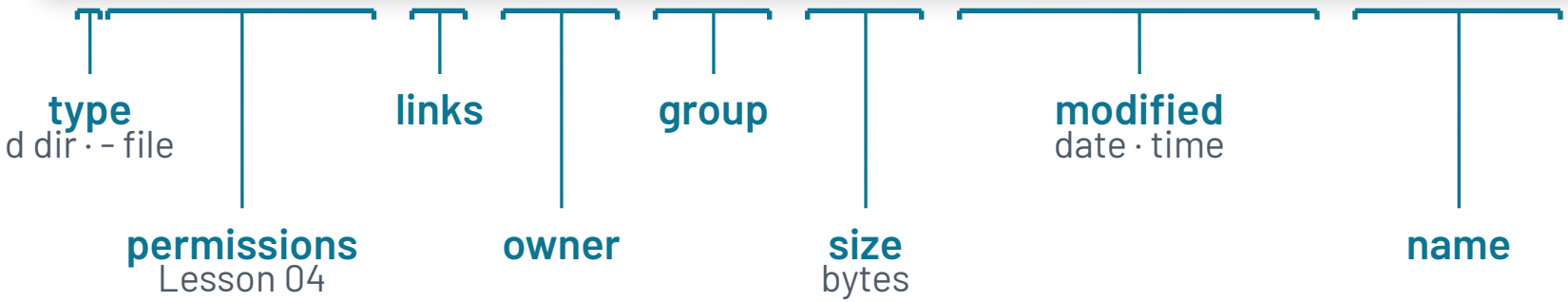
File and Directory Management

CYBERA4U

What is `ls -la` really telling you?

kali@kali: ~/lab

```
┌──(kali㉿kali)-[~/lab]
└─$ ls -la
total 12
drwxr-xr-x  2 kali kali 4096 Oct  2 09:14 .
drwx------ 18 kali kali 4096 Oct  2 09:10 ..
-rwxr-xr-x  1 kali kali  245 Oct  2 09:14 scan.sh
```



Move around

`cd /etc`
Jump to a path

`cd ..`
Up one level

`cd -`
Back to previous

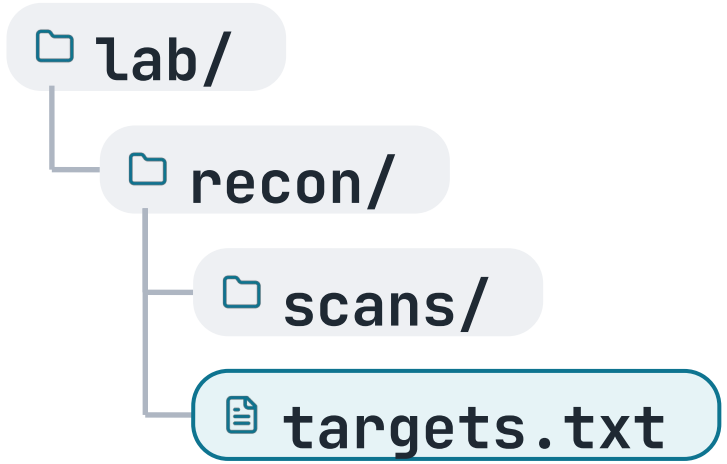
`cd ~`
Home again

CYBERA4U

Build a workspace in three commands

```
kali@kali: ~
┌──(kali㉿kali)-[~]
└─$ mkdir -p lab/recon/scans
┌──(kali㉿kali)-[~]
└─$ touch lab/recon/targets.txt
┌──(kali㉿kali)-[~]
└─$ nano lab/recon/targets.txt
```

RESULT



-p creates every missing parent

nano editor basics

Ctrl + **O** Save

Enter Confirm name

Ctrl + **X** Exit

CYBERA4U

Copy, move, rename, delete

`cp report.txt report.bak`
Copy a file

`cp -r lab/ lab-backup/`
Copy a folder (-r)

`mv draft.txt final.txt`
Move or rename

`rm notes.txt · rm -r old/`
Delete file or folder

`rm -rf <path>`

No recycle bin, no undo
Read the path twice

Match many files with one pattern

`*`	Any characters
`?`	One character
`[ ]`	One from the set

`ls *.txt` → `scan1.txt` `scan2.txt` `scan10.txt` ~~`scan1.log`~~ `Scan3.txt` ~~`notes.md`~~

`ls scan?.txt` → `scan1.txt` `scan2.txt` ~~`scan10.txt`~~ ~~`scan1.log`~~ ~~`Scan3.txt`~~ ~~`notes.md`~~

`ls scan[12].*` → `scan1.txt` `scan2.txt` ~~`scan10.txt`~~ `scan1.log` ~~`Scan3.txt`~~ ~~`notes.md`~~

CYBERA4U

find searches live; locate searches an index

FIND · LIVE SEARCH, ALWAYS CURRENT

```
find /etc -type f -name "*.conf"
```

- **command**
- **start here** — search root
- **file type** — f file · d dir
- **name pattern** — quote wildcards

```
find / -size +100M
```

Files over 100 MB

LOCATE · INSTANT, MAY BE STALE

```
kali@kali: ~
┌──(kali㉿kali)-[~]
└─$ sudo updatedb
┌──(kali㉿kali)-[~]
└─$ locate rockyou
/usr/share/wordlists/rockyou.txt.gz
```

Refresh the index first

updatedb rebuilds the file database

CYBERA4U

What is this command, and what is this file?

Which binary runs?

```
kali@kali: ~
┌──(kali㉿kali)-[~]
└─$ which nmap
/usr/bin/nmap
```

Where are its files?

```
kali@kali: ~
┌──(kali㉿kali)-[~]
└─$ whereis nmap
nmap: /usr/bin/nmap /usr/share/nmap /usr/share/man/man1/nmap.1.gz
```

What kind of file?

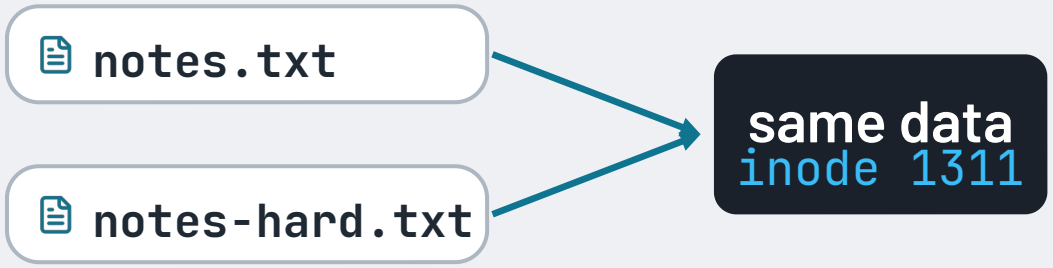
```
kali@kali: ~
┌──(kali㉿kali)-[~/lab]
└─$ file scan.sh
scan.sh: Bourne-Again shell script, ASCII text executable
```


CYBERA4U

Hard link or symbolic link?

HARD LINK

```
ln notes.txt notes-hard.txt
```



Delete one name, data survives

SYMBOLIC LINK

```
ln -s /usr/share/wordlists ~/wordlists
```



Points to a path; breaks if it moves

```
kali@kali: ~
┌──(kali㉿kali)-[~]
└─$ ls -l ~/wordlists
lrwxrwxrwx 1 kali kali 20 Oct  2 09:30 /home/kali/wordlists -> /usr/share/wordlists
```


CYBERA4U

Pack and unpack with tar, zip and gzip

```
tar -czvf lab.tar.gz lab/
```



Unpack

```
tar -xzvf lab.tar.gz
```

x = extract

zip / unzip

Cross-platform .zip archives

```
zip -r lab.zip lab/
```

```
unzip lab.zip
```

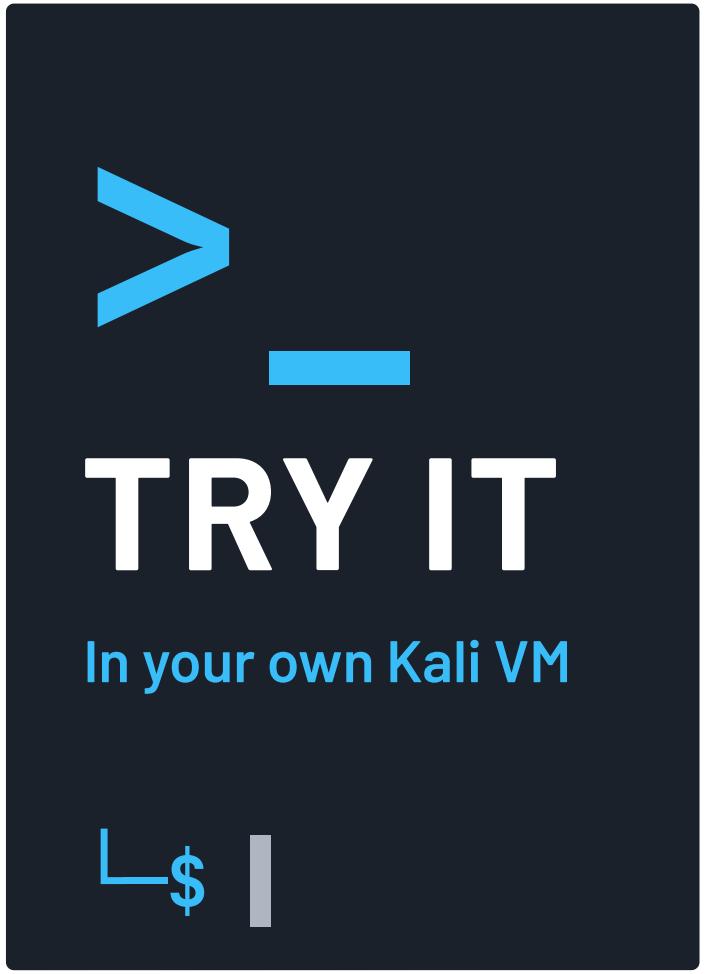
gzip / gunzip

Compress a single file

```
gzip notes.txt
```

```
gunzip rockyou.txt.gz
```

Try it: build and search a workspace



>01 Create ~/lab/recon/scans in one go.

>02 Find every .conf file under /etc.

>03 Symlink the wordlists folder into ~.

>04 Archive ~/lab, then extract it in /tmp.

CYBERA4U

What to remember from Lesson 03

+01

`ls -la` shows type, owner, size, time

+02

`rm` has no undo: read the path twice

+03

`find` searches live; `locate` searches an index

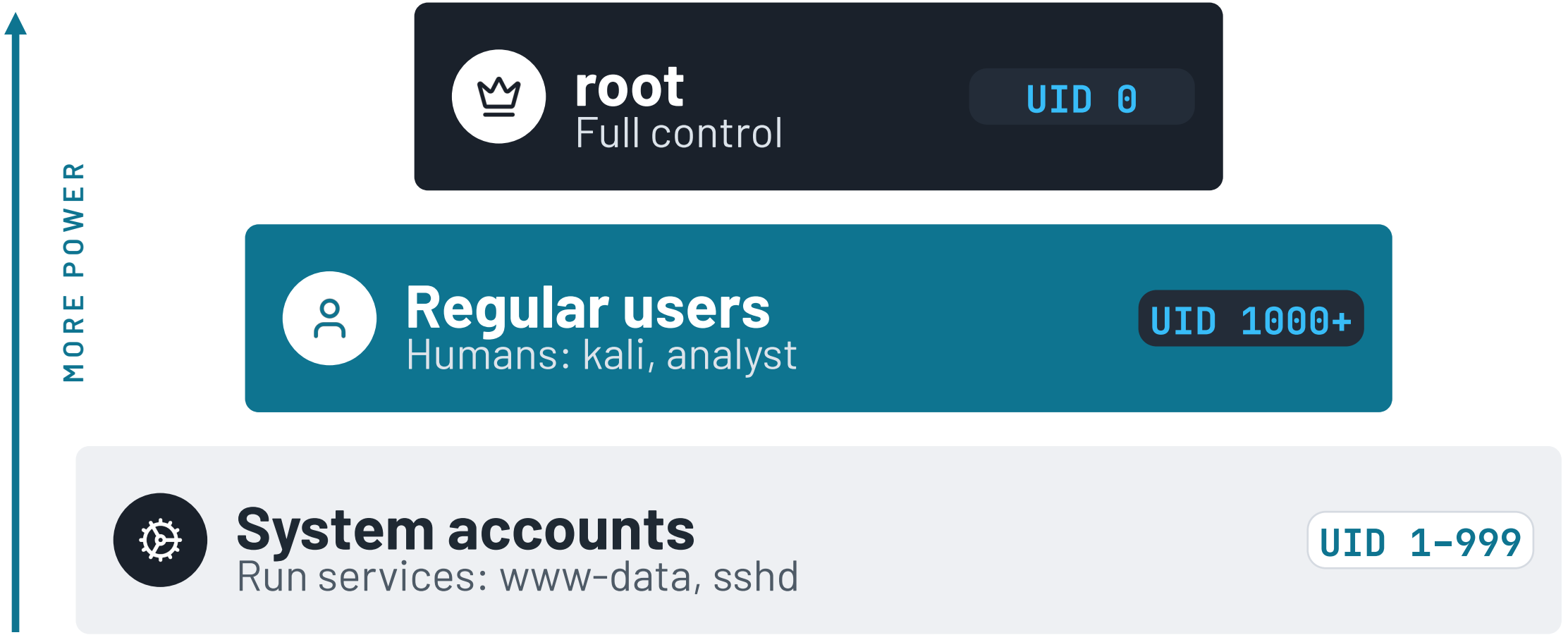
CYBERA4U

LESSON 04

Users, Groups and Permission Handling

CYBERA4U

Three kinds of accounts, three levels of power



CYBERA4U

What does one line of /etc/passwd say?

```
root:x:0:0:root:/root:/usr/bin/zsh
```



/etc/group

```
sudo:x:27:kali
```



/etc/shadow

Password hashes live here

Normal users cannot read it

```
-rw-r----- 1 root shadow
```


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Work as kali, borrow root only when needed

`sudo <cmd>`

- One command, then back
- Uses your password

`su - <user>`

- Whole shell switches user
- Kali root has no password

Daily work as a normal user

```
kali@kali: ~
┌──(kali㉿kali)-[~]
└─$ sudo whoami
[sudo] password for kali:
root
```

```
root@kali: ~
┌──(kali㉿kali)-[~]
└─$ sudo -i
┌──(root㉿kali)-[~]
└─# whoami
root
```

Create a user, then grant a group

1. **`sudo adduser analyst`**
   Create the account
2. **`sudo passwd analyst`**
   Set or change password
3. **`sudo usermod -aG wireshark analyst`**
   Append to a group
4. **`groups analyst`**
   Verify membership

```
kali@kali: ~
┌──(kali㉿kali)-[~]
└─$ groups analyst
analyst : analyst wireshark
```

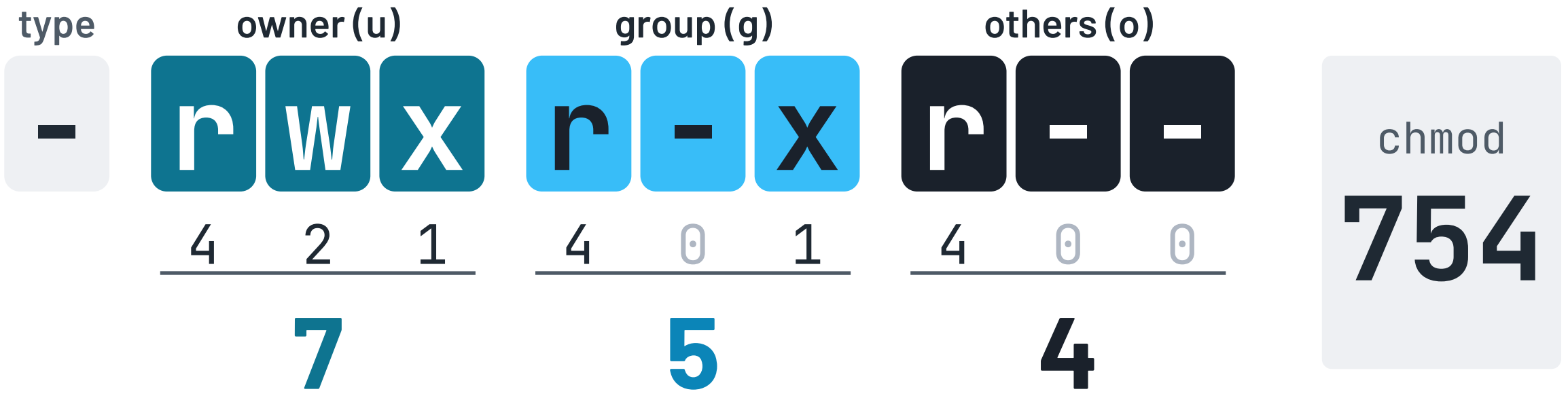
⚠ **`usermod -G`**

Without -a it replaces every group

`id` shows UID, GID and groups

CYBERA4U

How do you read `-rwxr-xr--`?



chmod **754**

`r` = **4** read

`w` = **2** write

`x` = **1** execute

Add the bits per group

CYBERA4U

Change permissions and ownership

chmod

```
chmod 640 notes.txt
```

Numeric: owner rw, group r

```
chmod +x scan.sh
```

Symbolic: add execute

```
chmod g-w report.txt
```

Symbolic: group loses write

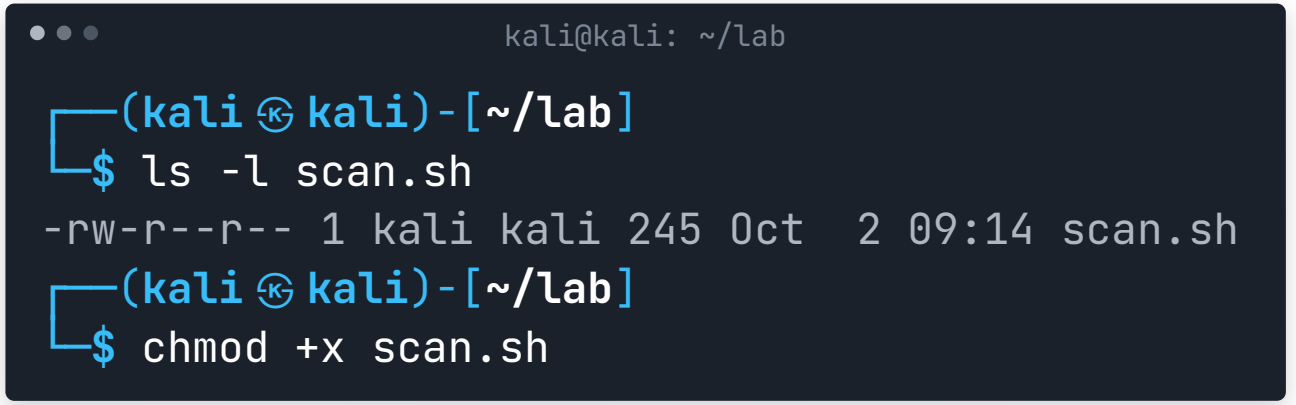
chown · chgrp

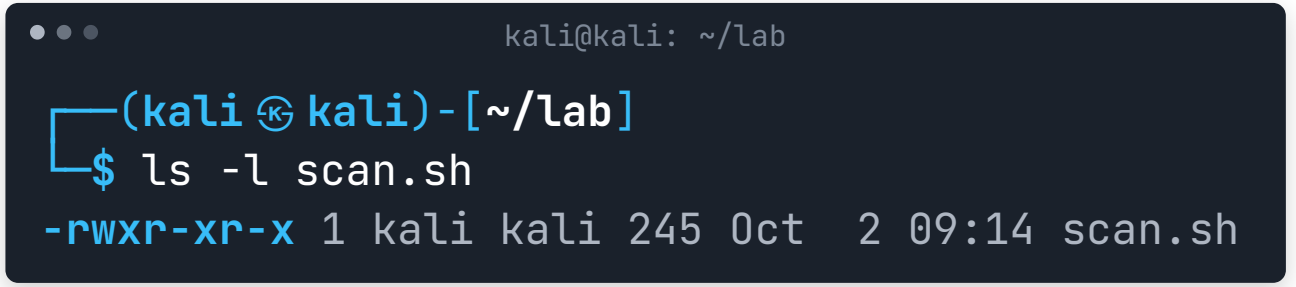
```
sudo chown kali:kali report.txt
```

New owner and group

```
sudo chgrp wireshark cap.pcap
```

New group only





Execute bit added for all three

CYBERA4U

Grant the least permission that works



chmod 777

rwxrwxrwx

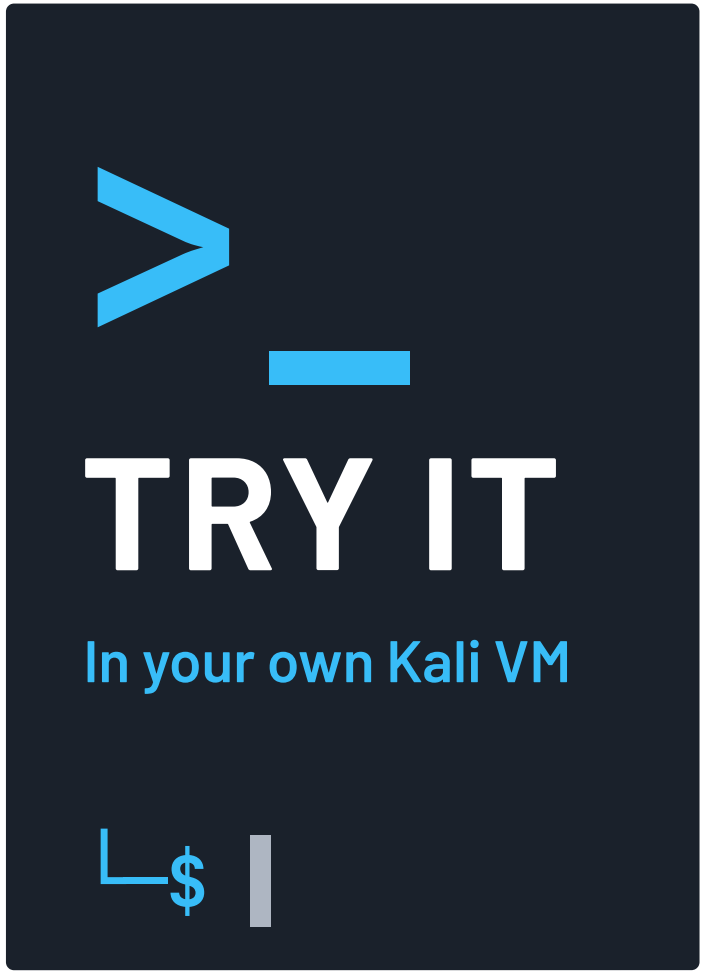
Never do this

Anyone can change or replace it

	Octal	Symbolic	Use it for
✓	755	rwxr-xr-x	Scripts, folders
✓	644	rw-r--r--	Regular files
✓	600	rw-------	Private keys, secrets
✓	700	rwx------	Private folders

Start strict; loosen only when something breaks

Try it: users and permissions



>01 Create user analyst with a password.

>02 Add analyst to the wireshark group.

>03 Make scan.sh executable for you only.

>04 Find which group owns /etc/shadow.

CYBERA4U

What to remember from Lesson 04

+01

Work as kali; `sudo` only when needed

+02

`r=4 w=2 x=1`: add per group

+03

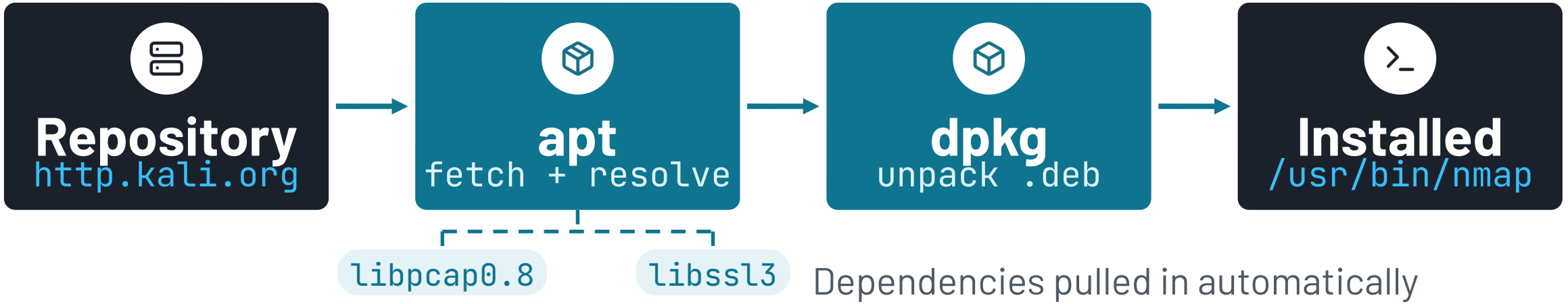
Never `777`: grant the least that works

LESSON 05

Software and Package Management

CYBERA4U

Where does Kali software come from?



/etc/apt/sources.list

```
deb http://http.kali.org/kali kali-rolling main contrib non-free non-free-firmware
```

type
binary packages

mirror
Kali's repository

kali-rolling
always-current branch

components
license sections

CYBERA4U

update, upgrade, full-upgrade: what changes?

`sudo apt update`

Refresh package lists
Installs nothing

`sudo apt upgrade`

Install newer versions
Never removes packages

`sudo apt full-upgrade`

Upgrade everything
May remove or replace
Kali's recommended upgrade

kali@kali: ~

```
┌──(kali㉿kali)-[~]
└─$ sudo apt update
Get:1 http://http.kali.org/kali kali-rolling InRelease [34.0 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 Packages [20.9 MB]
Get:3 http://http.kali.org/kali kali-rolling/main amd64 Contents (deb) [51.7 MB]
Fetched 72.6 MB in 9s (8,066 kB/s)
```


CYBERA4U

Find, install, remove: the apt lifecycle

FIND IT		GET IT	CLEAN UP		
apt **search**	apt **show**	apt **install**	apt **remove**	apt **purge**	apt **autoremove**
Find by keyword	Read the details	Add with dependencies	Delete, keep config	Delete with config	Drop unused deps

kali@kali: ~

```
┌──(kali㉿kali)-[~]
└─$ sudo apt install gobuster
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
```

apt pulls every dependency for you

apt or dpkg: which one should you use?

apt

Online, high level

- Downloads from repositories
- Resolves dependencies

```
sudo apt install nmap
```

dpkg

Local, low level

- Installs a local .deb
- No dependency resolution

```
sudo dpkg -i tool.deb
```

```
dpkg -l | grep nmap
```

Have a local .deb? Let apt fix dependencies

```
sudo apt install ./tool.deb
```


CYBERA4U

One package name, hundreds of tools

kali-linux-large
Default set plus extra tools



kali-tools-* · one category at a time

web · passwords · wireless · information-gathering · vulnerability · exploitation · sniffing-spoofing · forensics

```
kali@kali: ~
┌──(kali㉿kali)-[~]
└─$ sudo apt install kali-tools-web
```


CYBERA4U

Tool not packaged? Build it from source

1 `git clone <url>` — Get the code

2 `cat README.md` — Read install notes

3 `pipx install .` — Install in a venv

4 `<tool> --help` — Run and verify

pip install <tool>

```
kali@kali: ~
┌──(kali㉿kali)-[~]
└─$ pip install requests
error: externally-managed-environment
```

pipx install <tool>

Kali's Python is externally managed

One isolated venv per tool

```
sudo apt install pipx
```


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Start, stop and check services

```
sudo systemctl start apache2
```

- **manager** — systemd control
- **verb** — what to do
- **service** — apache2, ssh...

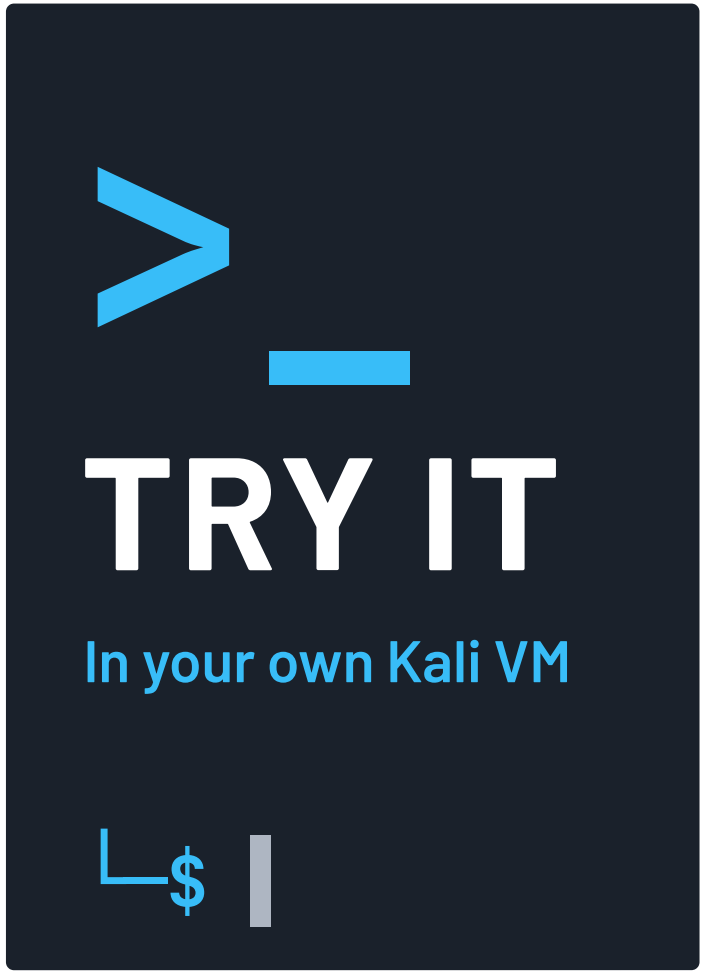
start — Run now	**stop** — Stop now
restart — Stop, then start	**status** — Is it running?
enable — Start at boot	**disable** — Not at boot

kali@kali: ~

```
┌──(kali㉿kali)-[~]
└─$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
     Active: active (running) since Thu 2026-10-01 09:12:44 EDT; 3min ago
```


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Try it: install and manage software



>01 Refresh lists; show what can upgrade.

>02 Show details of the gobuster package.

>03 Check if nmap is installed via dpkg.

>04 Start ssh, check it, stop it.

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What to remember from Lesson 05

+01

Always `apt update` before installing

+02

`apt` resolves dependencies; `dpkg` does not

+03

Python tools from source? Use `pipx`

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LESSON 06

Basic Networking Commands

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What is your lab identity and route?

legacy: ifconfig

kali@kali: ~

```
┌──(kali㉿kali)-[~]
└─$ ip a show eth0
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
 qlen 1000
    link/ether 08:00:27:3a:5c:1e brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0
```

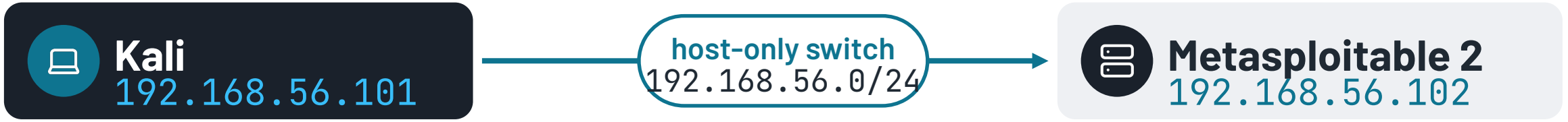
kali@kali: ~

```
┌──(kali㉿kali)-[~]
└─$ ip route
192.168.56.0/24 dev eth0 proto kernel scope link src 192.168.56.101 metric 100
```

No gateway
isolated

INTERFACE	STATE	MAC ADDRESS	IPv4 / CIDR
eth0	UP	08:00:27:3a:5c:1e	192.168.56.101/24

Can you reach the target?



kali@kali: ~

```
┌──(kali㉿kali)-[~]
└─$ ping -c 3 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=0.412 ms
```

-c 3
Stop after three replies

kali@kali: ~

```
┌──(kali㉿kali)-[~]
└─$ traceroute 192.168.56.102
traceroute to 192.168.56.102 (192.168.56.102), 30 hops max, 60 byte packets
 1  192.168.56.102 (192.168.56.102)  0.398 ms  0.371 ms  0.355 ms
```

1 hop
Same subnet, no router

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How does a name become an IP address?

metasploitable
Name to resolve

↓

/etc/hosts
Checked first

↓

/etc/resolv.conf
Lists DNS servers

↓

DNS server
Answers the query

/etc/hosts

```
127.0.0.1       localhost
127.0.1.1       kali
192.168.56.102  metasploitable      added for the lab
```

kali@kali: ~

```
┌──(kali㉿kali)-[~]
└─$ ping -c 1 metasploitable
PING metasploitable (192.168.56.102) 56(84) bytes of data.
```

dig **host** **nslookup**

Query DNS servers directly; they skip /etc/hosts

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What is listening on your machine?

```
kali@kali: ~

┌──(kali㉿kali)-[~]
└─$ sudo ss -tulpn
Netid State  Recv-Q Send-Q Local Address:Port Peer Address:Port Process
tcp   LISTEN 0      128          0.0.0.0:22        0.0.0.0:*    users:(("sshd",pid=842,fd=3))
tcp   LISTEN 0      128             [::]:22           [::]:*    users:(("sshd",pid=842,fd=4))
```

protocol
tcp or udp

state
LISTEN = waiting

local address:port
0.0.0.0 = all interfaces

process
who owns the port

```
sudo ss -tulpn
```

-t
TCP

-u
UDP

-l
listening

-p
process

-n
numbers

```
sudo netstat -tulpn
```

Same view, older tool
Needs the net-tools package

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Talk to a web server from the terminal

curl -I

`curl -I http://192.168.56.102`

Headers only: status, server

wget

`wget http://192.168.56.102/`

Download the page to a file

Metasploitable 2
192.168.56.102:80

kali@kali: ~

```
┌──(kali㉿kali)-[~]
└─$ curl -I http://192.168.56.102
HTTP/1.1 200 OK
Date: Thu, 01 Oct 2026 13:20:11 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
```

kali@kali: ~

```
┌──(kali㉿kali)-[~]
└─$ wget http://192.168.56.102/
--2026-10-01 09:20:11--  http://192.168.56.102/
Connecting to 192.168.56.102:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
```


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Move files inside the lab with a quick web server

kali@kali: ~/share · 192.168.56.101

```
┌──(kali㉿kali)-[~/share]
└─$ python3 -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
192.168.56.102 - - [01/Oct/2026 09:41:07] "GET /notes.txt HTTP/1.0" 200 -
```

Kali serves :8000
192.168.56.101

GET /notes.txt

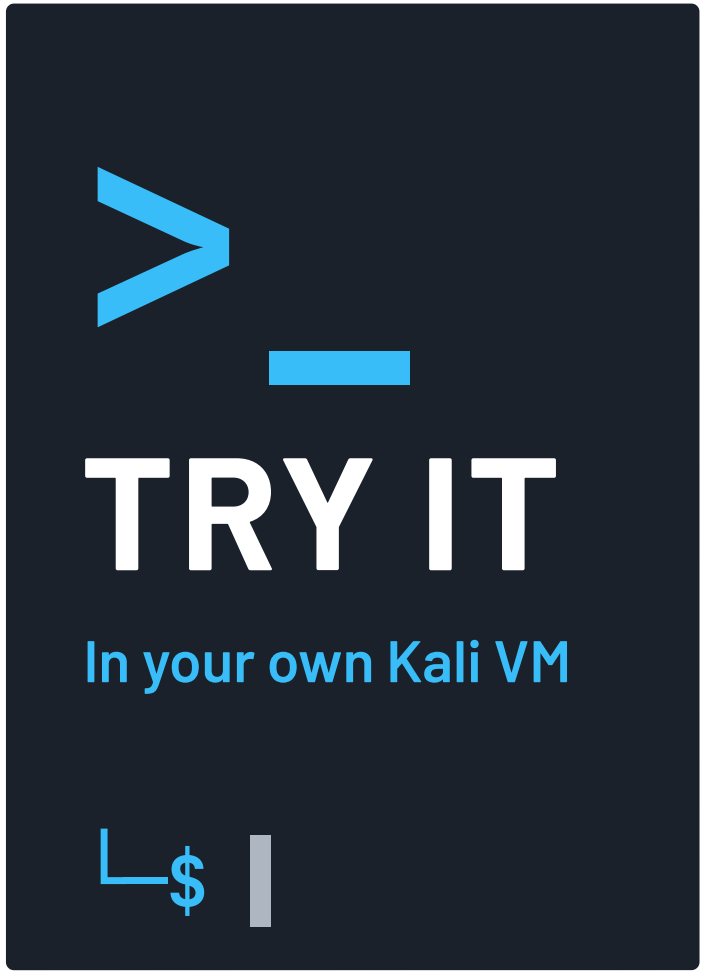
Metasploitable
192.168.56.102

msfadmin@metasploitable: ~

```
msfadmin@metasploitable:~$ wget http://192.168.56.101:8000/notes.txt
--09:41:07--  http://192.168.56.101:8000/notes.txt
          => `notes.txt'
```

Lab only
Isolated 192.168.56.0/24 only

Try it: map your lab network



>01 Find your Kali IP and MAC address.

>02 Ping Metasploitable exactly three times.

>03 List listening ports with their processes.

>04 Serve a file; fetch it from Metasploitable.

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What to remember from Lesson 06

+01

`ip a` and `ip route`: who and where

+02

`ss -tulpn` shows what your box exposes

+03

Practice only inside `192.168.56.0/24`

Cheat sheet: Lessons 01–03

Command	Purpose	Example
>01 Operating System File Structure Basics		
`pwd`	Current directory	`pwd`
`cd`	Change directory	`cd /etc`
`ls -a`	Show hidden files	`ls -a ~`
>02 Essential Terminal Commands		
`man`	Full manual	`man ls`
`whatis`	One-line summary	`whatis grep`
`apropos`	Keyword search	`apropos copy`
`--help`	Quick usage	`ls --help`
`whoami · id`	User and groups	`id`
`uname -a`	Kernel, system	`uname -a`
`cat · less`	Print or page	`less /etc/passwd`
`head · tail`	First, last lines	`tail -n 20 notes.txt`
`tail -f`	Follow a log	`tail -f app.log`
`grep`	Match lines	`grep -in error app.log`
`sort · uniq`	Order, count	`sort names | uniq -c`
`cut`	Pick fields	`cut -d: -f1 /etc/passwd`
`wc -l`	Count lines	`wc -l targets.txt`
`| > >>`	Pipe or redirect	`ls | grep txt > out`

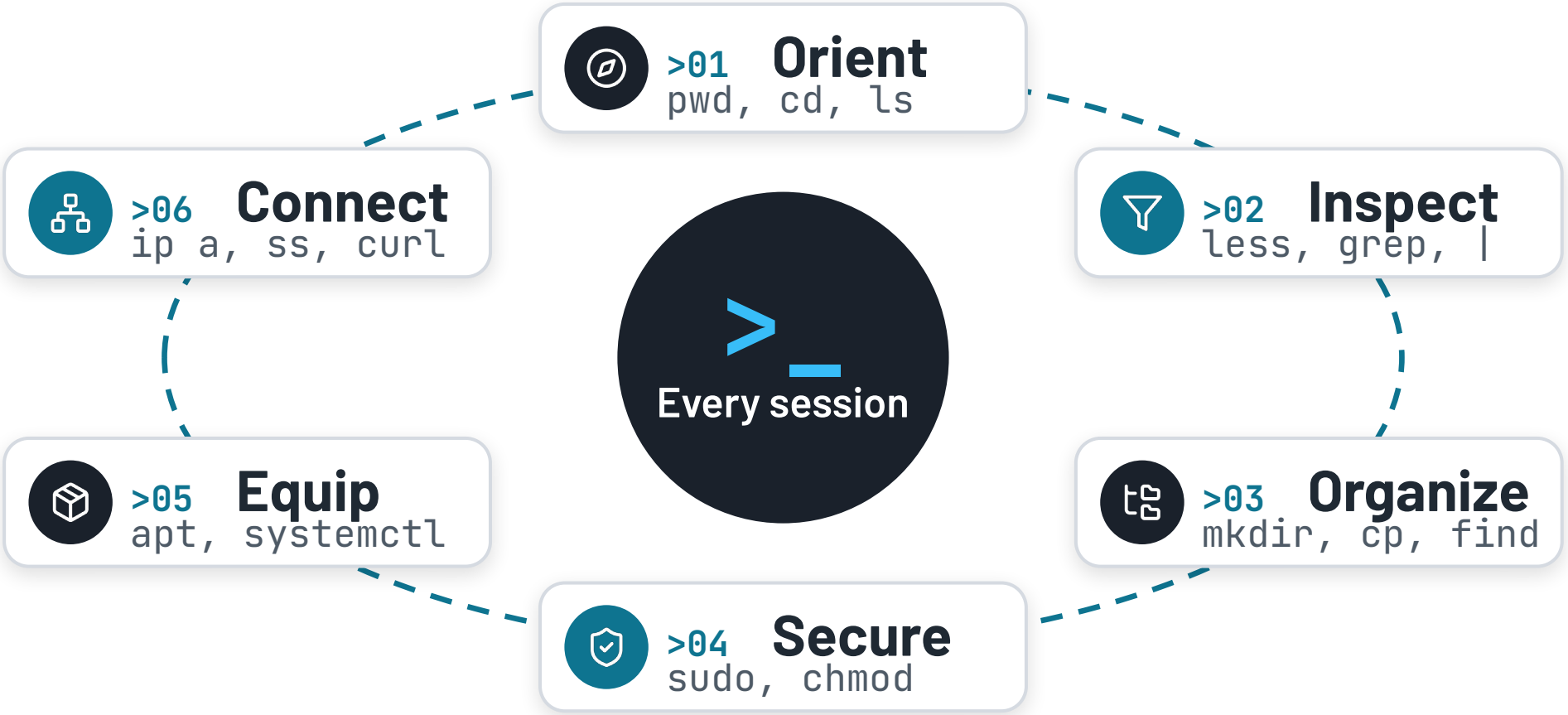
Command	Purpose	Example
>02 Essential Terminal Commands (cont.)		
`2>/dev/null`	Discard errors	`ls /root 2>/dev/null`
`history · !!`	Reuse history	`sudo !!`
>03 File and Directory Management		
`ls -la`	Long listing	`ls -la ~`
`cd - · cd ~`	Back, home	`cd -`
`mkdir -p`	Nested folders	`mkdir -p lab/recon`
`touch · nano`	Create, edit	`nano notes.txt`
`cp -r`	Copy	`cp -r lab lab.bak`
`mv`	Move, rename	`mv a.txt b.txt`
`rm · rm -r`	Delete, no undo	`rm -r old/`
`find`	Live search	`find / -name "*.conf"`
`locate`	Index search	`locate rockyou`
`updatedb`	Refresh index	`sudo updatedb`
`which/whereis`	Find a binary	`which nmap`
`file`	File type	`file scan.sh`
`ln -s`	Symbolic link	`ln -s target linkname`
`tar`	Archive, extract	`tar -czvf lab.tgz lab/`
`zip · unzip`	.zip archives	`unzip lab.zip`
`gzip · gunzip`	Compress a file	`gunzip notes.txt.gz`

Cheat sheet: Lessons 04–06

Command	Purpose	Example
>04 Users, Groups and Permission Handling		
`sudo`	Run as root	`sudo apt update`
`sudo -i`	Root shell	`sudo -i`
`su -`	Switch user	`su - analyst`
`adduser`	Create a user	`sudo adduser analyst`
`passwd`	Set password	`sudo passwd analyst`
`usermod -aG`	Add to group	`sudo usermod -aG sudo bob`
`groups`	Memberships	`groups analyst`
`chmod`	Permissions	`chmod 755 scan.sh`
`chown`	Owner	`sudo chown kali f.txt`
`chgrp`	Group	`sudo chgrp staff f.txt`
>05 Software and Package Management		
`apt update`	Refresh lists	`sudo apt update`
`apt upgrade`	Safe upgrade	`sudo apt upgrade`
`apt full-upgrade`	Full upgrade	`sudo apt full-upgrade`
`apt install`	Install	`sudo apt install gobuster`
`apt purge`	Remove, config	`sudo apt purge gobuster`

Command	Purpose	Example
>05 Software and Package Management (cont.)		
`apt autoremove`	Unused deps	`sudo apt autoremove`
`apt search/show`	Find, inspect	`apt show gobuster`
`dpkg -i · -l`	Local .deb, list	`sudo dpkg -i tool.deb`
`git clone`	Get source	`git clone <url>`
`pipx install`	Python tools	`pipx install .`
`systemctl`	Services	`sudo systemctl start ssh`
>06 Basic Networking Commands		
`ip a`	Address, MAC	`ip a show eth0`
`ip route`	Routes	`ip route`
`ping -c`	Reachability	`ping -c 3 192.168.56.102`
`traceroute`	Path to host	`traceroute 192.168.56.102`
`dig · host`	DNS lookup	`dig <name>`
`ss -tulpn`	Listening ports	`sudo ss -tulpn`
`netstat -tulpn`	Legacy ports	`sudo netstat -tulpn`
`curl -I`	HTTP headers	`curl -I 192.168.56.102`
`wget`	Download	`wget 192.168.56.102/`
`http.server`	Share a folder	`python3 -m http.server`

Your daily terminal workflow

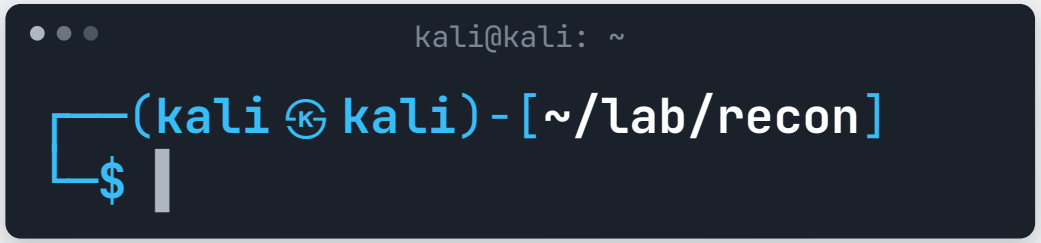


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Knowledge check: questions 1–2

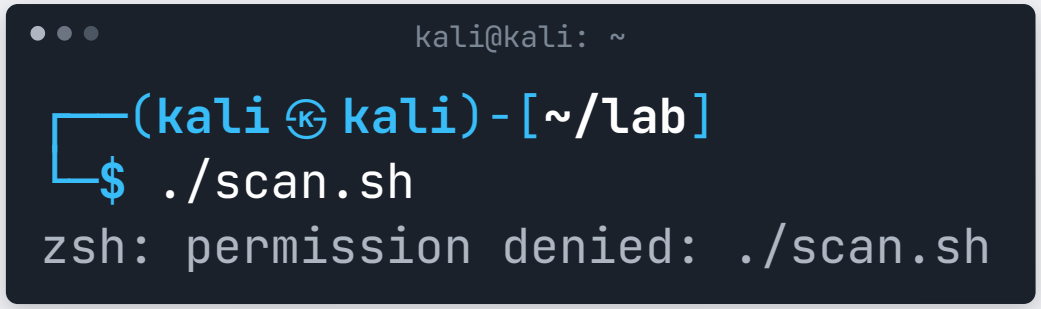
Q1

Which command prints your current directory?



Q2

./scan.sh says “permission denied”. How do you make it executable?



Knowledge check: questions 3–4

Q3

Which command finds “password” in config.php, ignoring case?



`config.php`

Q4

Which command lists listening ports and their owning processes?



`Kali · 192.168.56.101`

Knowledge check: question 5

Q5

gobuster is not installed. Which two commands install it, in order?

```
1 └─$ ____________________________
2 └─$ ____________________________
```



PROFESSIONAL ETHICAL HACKING TRACK · PEHT

Next: Module 04

INSTRUCTOR

Mo'tasem Abu Baha'

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